

Design Reflection: Mini Calculator

1. Main Functionalities

This report describes the design and implementation of the Mini Calculator application. It is built using Java and the Swing library. The application can perform basic arithmetic operations, including addition (+), subtraction (-), multiplication (*), division (/), and modulo (%). It also includes control buttons. Users can clear all input using the "C" button or delete the last character using the "<-" button. The result is clearly displayed on the screen.

2. Design Decisions & Rationale

The core design decision for this assignment is the Separation of Concerns principle. The code is divided into three main classes: Main, CalculatorGUI, and CalculatorLogic. The Main class starts the application. The CalculatorGUI class manages the user interface and handles button click events by implementing ActionListener. The CalculatorLogic class handles arithmetic calculations and returns the result. This separation makes the code cleaner, easier to read, easier to test, and easier to fix when bugs occur.

3. Brief Explanation of the Implementation

For the user interface layout, the application combines BorderLayout and GridLayout. The JTextField display is placed at BorderLayout.NORTH so users can easily see the input and result. The buttons are organized in a JPanel using GridLayout(5, 4). This makes all buttons have the same size and keeps the interface neat. The program also handles invalid input safely. For example, if a user tries to divide by zero, the logic class throws an ArithmeticException. The GUI catches this error and shows "Error: divide by 0" on the screen. This prevents the application from crashing.

